UE24CS342AA2
Data Analytics
Unit 4 Worksheet Guide

There are two parts for the Unit 4 worksheet: a numerical section and a coding section.

Worksheet 4A:

This lab shows how a retailer can find items that are bought together and turn those patterns into association rules, using the Apriori algorithm on a small café dataset.

Students are to solve the worksheet using the dataset mentioned, and to only refer to their notes.

The numerical solution **must** be hand-written. Typed solutions will not be considered.

Worksheet 4A at a Glance:

• Part A (Q1 to Q4): Count supports and uses Apriori to find the frequent 1, 2 and 3 itemsets at min_sup = 30%.

• Part B (Q5): Turn the frequent 3-itemset into association rules and use confidence (min_conf = 70%) to keep only the strong ones.

• Bonus (Q6): Check whether a strong rule is actually useful by comparing its confidence with how popular the item is on its own.

Worksheet 4B:

In the coding section, you act as a Data Analyst intern at TalentMatch, a job portal. The company wants to go beyond keyword search and build a recommender that suggests job postings matching a candidate's skills, experience, location and job-type preferences.

You will build a content-based filtering system:

- Items are job postings.
- The "user" is a job seeker with a desired profile.
- Matching is done by comparing each posting's attributes (skills, industry, location, job type) against the candidate's profile using TF-IDF vectors and cosine similarity.

Dataset:

job_postings.json

Column	Meaning
Job Title	Role title (e.g. Data Analyst)
Company	Hiring company
Industry	Sector (Technology, Healthcare, Finance…)
Location	Region (North America, Europe, Asia…)
Experience Level	Years of experience required (integer, 0–15)
Skills	Comma-separated skill keywords (Python, SQL, ML…)
Job Type	Full-time, Part-time, Contract, Internship
Description	Short text description

How to Access This Notebook?

1. Open the Kaggle notebook shared with you.
2. Click **Copy & Edit** to create your own editable copy.
3. Confirm that the provided datasets are available under the notebook's Input section.
4. Review the imports and execute the notebook cells in the appropriate order.
5. Complete all TODO sections directly in the notebook.
6. Write your explanations and conclusions in the provided Markdown cells.

Use **Python** as your programming language. Do not replace the required implementation with a different approach simply because a ready-made package is available.

Worksheet 4B At A Glance:

- Problem 1: Feature Engineering: Build a feature_soup column combining Location, Industry, Skills and Job Type into one string per posting.
- Problem 2: Building the Recommender: Fit a TF-IDF vectorizer on feature_soup, filter jobs to 2-5 years of experience, then use cosine similarity to return the top 5 matches for the given candidate profile.
- Problem 3: Encapsulating the Logic: Wrap the whole pipeline into get_recommendations(candidate_profile, df, n=5) that returns the top n jobs.
- Problem 4: The Long-Tail Problem: Write a short answer on how a few very common skills and many rare ones affect TF-IDF weighting.

- Problem 5: Shortcomings and Improvements: Write the limitations of this approach and propose one improvement.

Submission Instructions:

Worksheet 4a:

Take pictures of your solution of the numerical and convert it to a pdf format. Use the format mentioned below to submit.

Worksheet 4b:

- Download your completed notebook using
  File → Download → Download Notebook.
- Ensure that all code cells have been executed and their outputs are visible.
- Complete all required Markdown explanations, interpretations, and conclusions.
- Submit the notebook using the Google Form link provided in the email.

Important Rules:

- To solve 4a, students can only refer to their notes.
- No devices are allowed to be used during the numerical section.
- Submissions for 4a will only be accepted if it is hand-written with steps.
- Complete the analysis of 4b in Python only.
- Do not leave required TODO sections or inference cells incomplete in 4b.
- Do not copy or share notebooks or solutions with other students. Copying solutions will result in no marks being awarded for the worksheet.
- Submissions after the deadline will not be accepted without prior approval.

Note: There may be multiple valid modelling approaches. Marks will be awarded for sound analytical reasoning, correct implementation and meaningful interpretation even if the modelling approach is not the exact same as the expected solution.

Please reach out to the TAs or faculty if you have any questions or encounter issues with the notebook.

Happy Learning :))